

The empirical recurrence for OEIS A282526: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A282526 records “Empirical recurrence of order 68”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 106$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282526 is “Number of nX6 0..1 arrays with no 1 equal to more than two of its king-move neighbors.”. It has offset 1 and begins

64, 1529, 29433, 747184, 17100576, 398640091, 9355397095, 218250094269, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Feb 17 2017 10:19:08, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1593$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 106$ of the 1593, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 106$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 212$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	20	c_{18}	−12824291315	c_{35}	−4905633615524	c_{52}	84712379894
c_2	91	c_{19}	44540747478	c_{36}	5547188999924	c_{53}	−51982136403
c_3	391	c_{20}	−221938450850	c_{37}	−2073505472742	c_{54}	−19062857803
c_4	−16344	c_{21}	125361528114	c_{38}	6453605321537	c_{55}	16056540140
c_5	16269	c_{22}	−88802088588	c_{39}	−623155428128	c_{56}	−2805996764
c_6	−150093	c_{23}	56613680668	c_{40}	127957219144	c_{57}	576024875
c_7	2093017	c_{24}	−465680371733	c_{41}	7525463936259	c_{58}	250545144
c_8	−2996640	c_{25}	1396195380650	c_{42}	−3448005019370	c_{59}	352015333
c_9	16163708	c_{26}	1732391603405	c_{43}	931428217400	c_{60}	366652444
c_{10}	−7402096	c_{27}	−1847320547265	c_{44}	2825069856343	c_{61}	−21897612
c_{11}	−133557617	c_{28}	−457440501485	c_{45}	−2061152594146	c_{62}	16080078
c_{12}	69378568	c_{29}	−856913836686	c_{46}	184836886790	c_{63}	2472507
c_{13}	−947945140	c_{30}	4301148824633	c_{47}	386660289005	c_{64}	−973648
c_{14}	1641121633	c_{31}	−5721287846727	c_{48}	−769421894786	c_{65}	572341
c_{15}	−889361367	c_{32}	−8276596518006	c_{49}	141171462935	c_{66}	−474434
c_{16}	12040681358	c_{33}	8244023940086	c_{50}	−122501342724	c_{67}	−4688
c_{17}	−1673613682	c_{34}	−3287886046518	c_{51}	−178734328133	c_{68}	568

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68 + 4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

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